EIU

Machine Learning on AWS



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Learning Objectives

01

Understand AWS ML Portfolio

Navigate the three-tier architecture of AWS AI and ML services, from pre-trained APIs to custom infrastructure.

02

Deploy ML Models in Production

Learn to build, train, and deploy machine learning models using Amazon SageMaker and related services.

03

Design Cloud-Native ML Systems

Apply architectural patterns for batch processing, real-time inference, and streaming predictions.

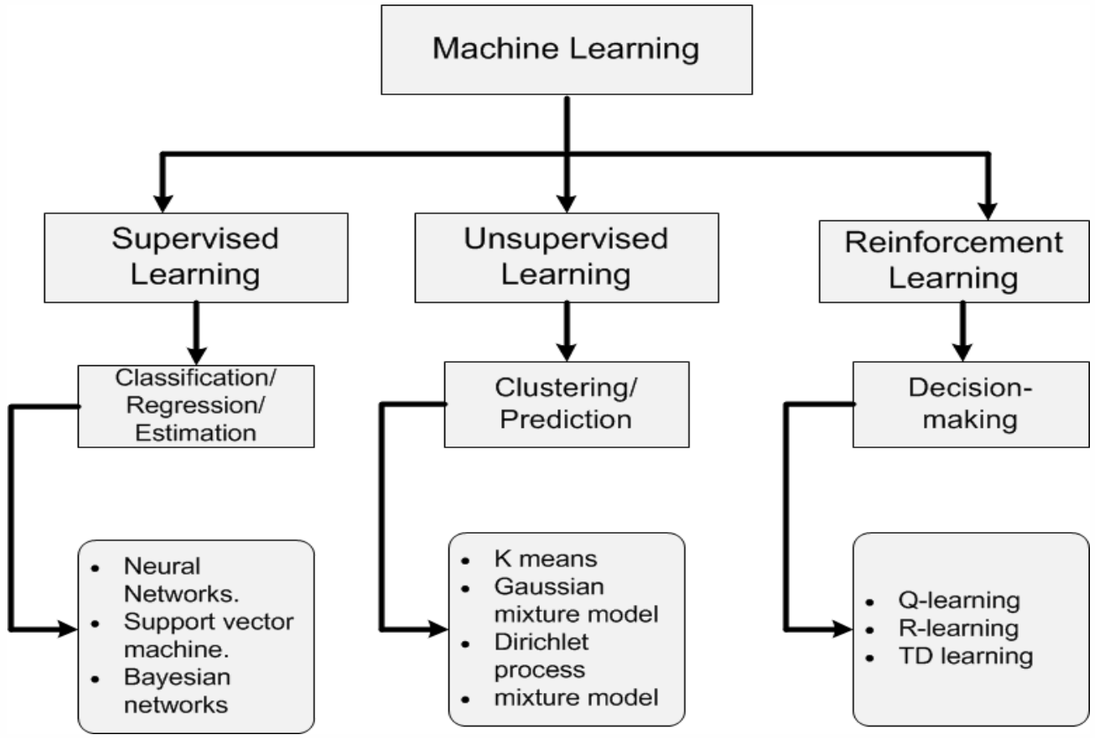
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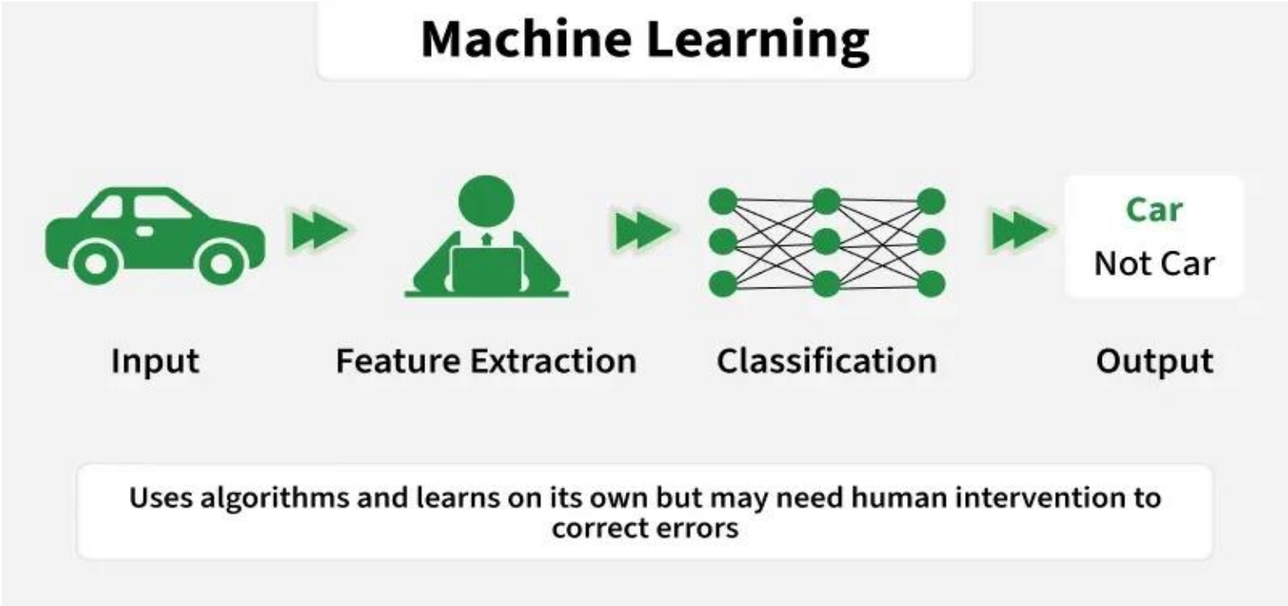
Optimize Costs and Security

Implement strategies for cost-effective ML operations while maintaining security and governance standards.

What is Machine Learning?

- A branch of Artificial Intelligence (AI)
- Enables systems to learn from data without explicit programming
- Improves predictions and decision-making over time





Common ML Types

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

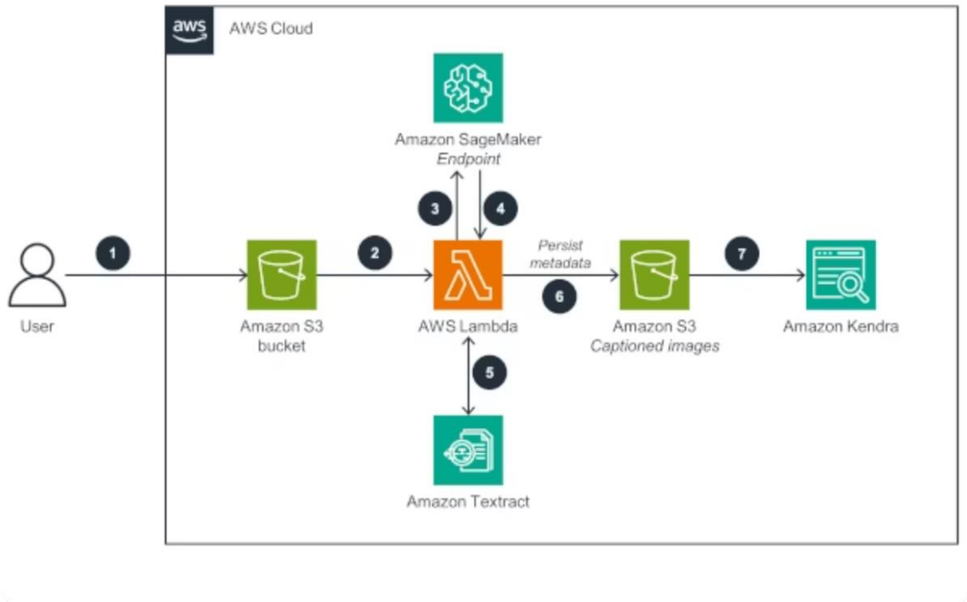
Why Machine Learning in the Cloud?

On-Premises Challenges

- High upfront hardware costs
- GPU procurement delays
- Capacity planning difficulties
- Maintenance overhead
- Limited scalability

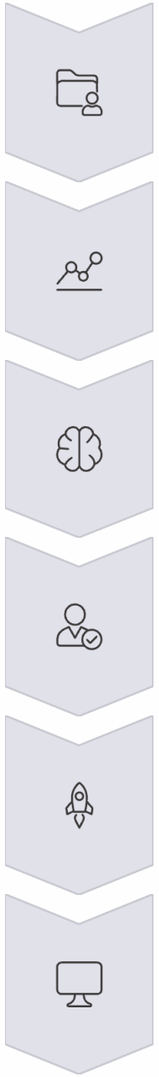
Cloud Advantages

- Pay-as-you-go pricing model
- Instant access to GPUs and TPUs
- Elastic scaling for training and inference
- Managed services reduce DevOps burden
- Global deployment capabilities



Cloud platforms democratize machine learning by removing infrastructure barriers, allowing teams to focus on model development rather than hardware management.

The Machine Learning pipeline



Data Collection

Gather and store training data in S3, databases, or data lakes

Data Preparation

Clean, transform, and feature engineer using Glue or EMR

Model Training

Train models using SageMaker with distributed computing

Model Evaluation

Validate performance and tune hyperparameters

Deployment

Serve predictions via endpoints or batch jobs

Monitoring

Track model drift and performance degradation

Importance of Data Preparation

- High-quality data improves model accuracy
- Reduces noise and inconsistencies

Data Preparation Tasks

- Data cleaning
- Handling missing values
- Data normalization
- Label encoding
- Splitting datasets

AWS Services for Data Prep

- Amazon S3
- AWS Glue
- Amazon EMR
- SageMaker Data Wrangler

What is Feature Engineering?

- Process of selecting and transforming data features
- Helps models learn patterns effectively

Common Techniques

- Feature scaling
- Feature extraction
- One-hot encoding
- Dimensionality reduction

AWS Support

- SageMaker Feature Store
- Automated feature processing tools

How Cloud Supports Each ML Phase

Data Phase

- **S3:** Unlimited storage for datasets
- **Glue:** ETL and data cataloging
- **Athena:** SQL queries on data lakes
- **Lake Formation:** Secure data lakes

Training Phase

- **SageMaker:** Managed training infrastructure
- **EC2 with GPUs:** Custom compute environments
- **Spot Instances:** 70-90% cost savings
- **Distributed Training:** Multi-node clusters

Deployment Phase

- **Real-time Endpoints:** Low-latency predictions
- **Lambda:** Serverless inference
- **Batch Transform:** Large-scale scoring
- **Edge Deployment:** IoT and mobile devices

Cloud services provide turnkey solutions for each phase, allowing teams to assemble ML pipelines from managed components rather than building everything from scratch.

Cloud ML: Pros and Cons

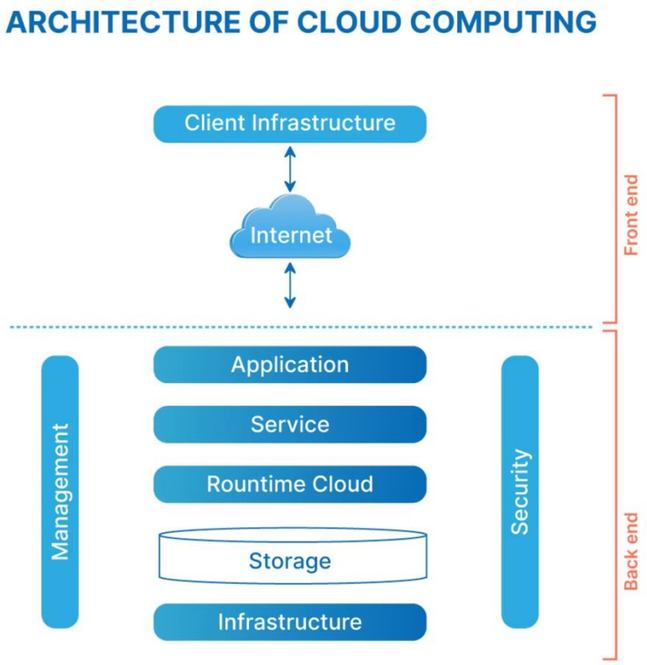
✓ Advantages

- Rapid experimentation with elastic resources
- Access to cutting-edge hardware (A100, H100 GPUs)
- Pre-built algorithms and frameworks
- Built-in MLOps and monitoring tools
- Global distribution and edge deployment
- Integration with enterprise systems

✗ Challenges

- Ongoing costs can exceed on-prem for sustained workloads
- Data transfer costs and latency considerations
- Vendor lock-in with proprietary services
- Learning curve for cloud-native architectures
- Compliance and data sovereignty concerns
- Cost management requires discipline

AWS AI & ML Portfolio: Three-Layer Architecture



ML Frameworks & Infrastructure

Bottom Layer: EC2 instances with GPUs, Deep Learning AMIs, EKS for Kubernetes, custom containers, and direct framework access (TensorFlow, PyTorch, MXNet). Maximum flexibility, maximum responsibility.

ML Services

Middle Layer: Amazon SageMaker provides end-to-end ML workflow management. Build, train, and deploy custom models with managed infrastructure. Balances control with convenience.

AI Services

Top Layer: Pre-trained APIs for vision, language, speech, and domain-specific tasks. No ML expertise required—just API calls. Fastest time to value for common use cases.

AI Services Layer: Pre-Trained APIs

AWS AI Services provide ready-to-use machine learning capabilities through simple API calls. These services are ideal when your use case matches common patterns and you need quick deployment without custom model training.



Vision

Image and video analysis, facial recognition, object detection



Language

Text analysis, translation, sentiment detection, entity extraction



Speech

Speech-to-text, text-to-speech, conversational interfaces



Documents

OCR, form processing, document structure analysis



Generative AI

Foundation models for text, images, and code generation

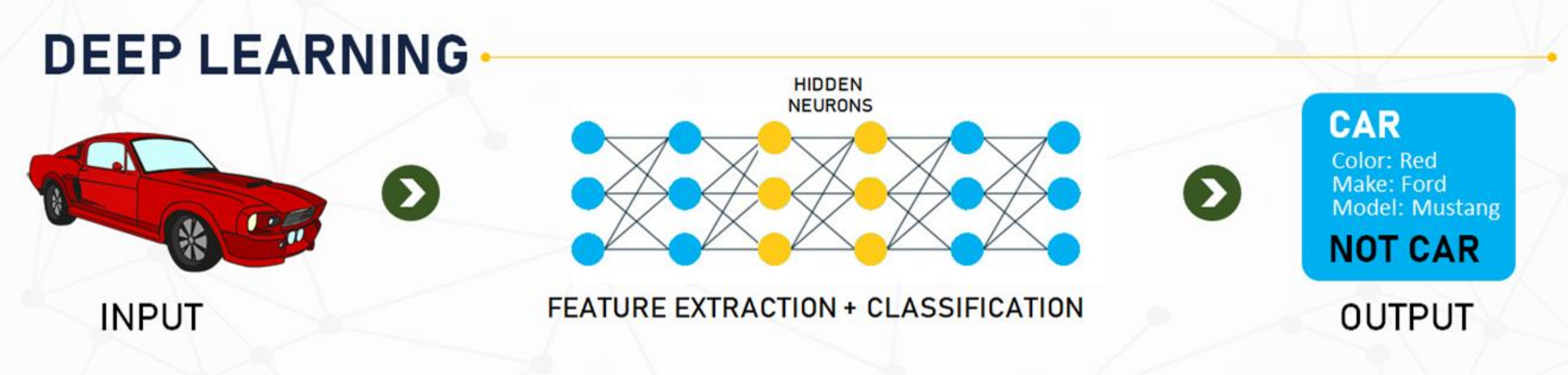


Domain-Specific

Recommendations, forecasting, fraud detection, personalization

What is Deep Learning?

- Advanced ML using neural networks
- Handles large and complex datasets



AWS Deep Learning Services

- SageMaker Deep Learning Containers
- AWS Inferentia
- AWS Trainium
- EC2 GPU instances

Amazon Rekognition: Computer Vision Made Simple

Core Capabilities

- **Object Detection:** Identify thousands of objects and scenes
- **Facial Analysis:** Detect faces and analyze attributes
- **Celebrity Recognition:** Identify public figures
- **Text in Images:** Extract text from photos
- **Content Moderation:** Flag inappropriate content
- **Video Analysis:** Track objects and people across frames



amazon Rekognition

Real-world scenarios: Content moderation for social media, identity verification for banking apps, visual search for e-commerce, security monitoring for smart buildings. **When NOT to use:** Highly specialized domains (medical imaging, satellite analysis), scenarios requiring fine-grained custom labels, or when training data differs significantly from Rekognition's training corpus.

Rekognition Response Example


```
{  "Labels": [    {      "Name": "Person",      "Confidence": 99.8,      "Instances": [        {          "BoundingBox": {            "Width": 0.23,            "Height":
0.67,            "Left": 0.42,            "Top": 0.15          },          "Confidence": 99.5        }      ]    },    {      "Name": "Laptop",      "Confidence": 98.2,
"Parents": [{"Name": "Electronics"}]    }  ]}
```


Rekognition returns structured JSON with detected labels, confidence scores, and bounding boxes for object localization. Simple API integration makes it accessible without deep ML knowledge.

What is NLP?

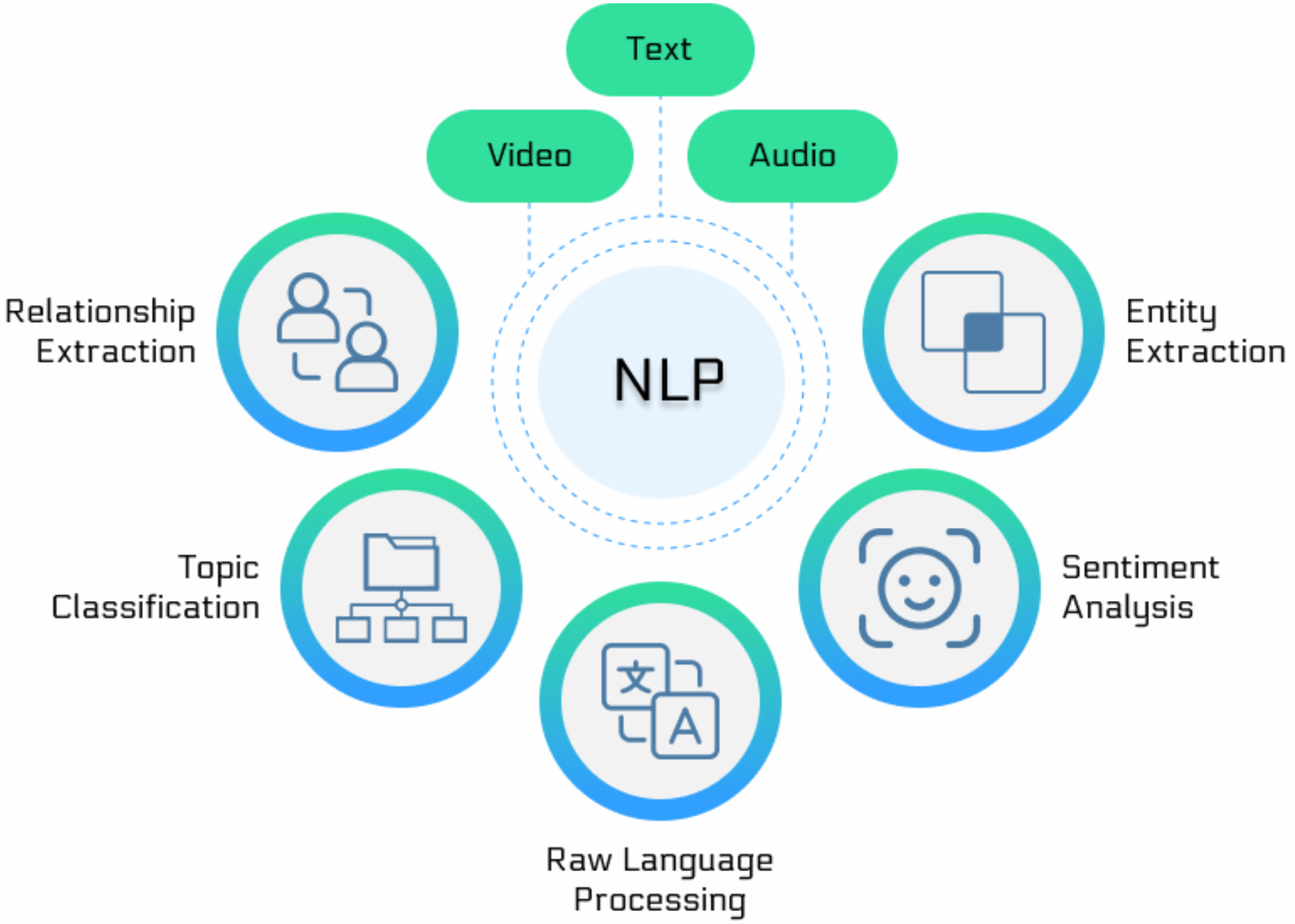
- Enables machines to understand human language

AWS NLP Capabilities

- Sentiment analysis
- Language translation
- Chatbots
- Speech recognition

Benefits

- Improved customer interaction
- Automated text analysis
- Faster communication



Language Services: Comprehend and Translate

Amazon Comprehend

Natural Language Processing: Extract insights from text including sentiment analysis, entity recognition, key phrase extraction, language detection, and topic modeling.

- Sentiment: Positive, negative, neutral, mixed
- Entities: People, places, organizations, dates
- Custom entity recognition for domain terms
- PII detection for privacy compliance

Use cases: Customer feedback analysis, document classification, social media monitoring, compliance screening.

Amazon Translate

Neural Machine Translation: Real-time translation supporting 75+ languages with automatic language detection and custom terminology support.

- Batch translation for large document sets
- Active Custom Translation for brand names
- Formality settings for target language
- Profanity masking capabilities

Use cases: Website localization, real-time chat translation, document translation, multilingual customer support.

Speech and Conversational AI Services



Amazon Transcribe

Automatic speech recognition that converts audio to text. Supports speaker identification, custom vocabularies, and real-time streaming transcription.



Amazon Polly

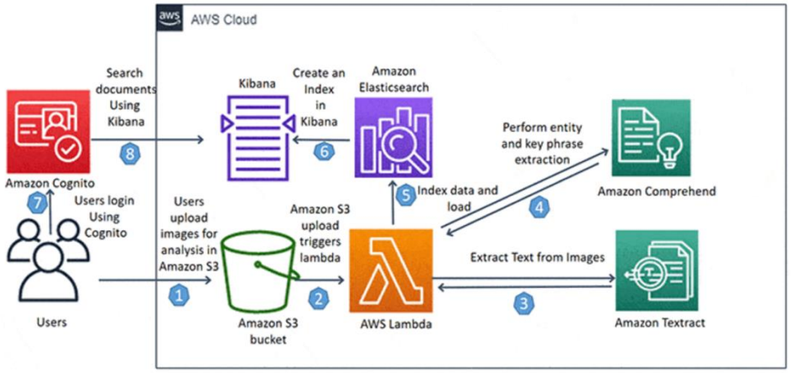
Text-to-speech service with lifelike voices in 60+ languages. Neural TTS produces highly natural speech with SSML support for pronunciation control.



Amazon Lex

Build conversational interfaces using the same technology as Alexa. Create chatbots with natural language understanding and multi-turn conversations.

Combined use case: Build an IVR system where Lex handles conversation flow, Transcribe converts caller speech to text, and Polly provides voice responses. Or create an accessible application where Transcribe makes audio searchable and Polly reads content aloud.



Amazon Textract: Document Intelligence

Amazon Textract goes beyond simple OCR by understanding document structure and extracting data from forms and tables automatically.

Text Detection

Extract printed and handwritten text from scanned documents, preserving layout and formatting.

Form Extraction

Identify key-value pairs in forms without custom templates. Automatically understands fields like "Name:", "Date:", "Amount:".

Table Extraction

Maintain table structure when extracting data, preserving rows, columns, and cell relationships.

Queries

Ask natural language questions about documents: "What is the invoice total?" or "Who is the vendor?"

Use cases: Invoice processing, loan application automation, medical records digitization, compliance documentation. **When NOT to use:** Complex scientific notation, non-standard languages, or when 100% accuracy is required without human review.

Domain-Specific AI Services

Amazon Personalize

Real-time personalization and recommendation engine using the same technology as Amazon.com. Feed user interaction data and get personalized product recommendations, search results, or content rankings.

- User-personalization recipes
- Similar-items recommendations
- Personalized ranking
- Real-time updates as users interact

Amazon Forecast

Time-series forecasting using machine learning to predict future values based on historical data. More accurate than traditional statistical methods.

- Demand forecasting for inventory
- Resource capacity planning
- Financial planning and budgeting
- Automated model selection

Amazon Fraud Detector: Identify potentially fraudulent online activities using ML models trained on AWS fraud detection expertise. Detects online payment fraud, fake account creation, and account takeover attempts.

Amazon Bedrock: Foundation Models as a Service

Amazon Bedrock provides access to high-performing foundation models from leading AI companies through a unified API, enabling developers to build generative AI applications without managing infrastructure.

Text Generation

Claude, Llama, Titan models for content creation, summarization, Q&A, and code generation

Image Generation

Stable Diffusion models for creating custom images from text descriptions

Conversational AI

Build sophisticated chatbots and virtual assistants with context awareness

Key features: Retrieval Augmented Generation (RAG) for grounding responses in your data, fine-tuning with your datasets, guardrails for safety, and knowledge bases for domain-specific applications. Pricing is pay-per-token with no minimum commitments.

When to Use AI Services vs. Custom Models

Choose AI Services When

- Your use case matches common patterns (face detection, sentiment analysis, object recognition)
- You need rapid deployment with minimal ML expertise
- You don't have large labeled training datasets
- Pay-per-request pricing fits your budget and scale
- Pre-trained models deliver acceptable accuracy
- You want AWS to handle model updates and improvements

Build Custom Models When

- You have specialized domain requirements (medical imaging, scientific analysis)
- Pre-trained models don't achieve required accuracy
- You have proprietary labeled training data
- Your data distribution differs significantly from public datasets
- You need full control over model behavior and updates
- High request volumes make custom endpoints more cost-effective

Amazon SageMaker

The complete platform for building, training, and deploying machine learning models at scale



What is Amazon SageMaker?

Amazon SageMaker is a fully managed machine learning platform that provides every developer and data scientist with the ability to build, train, and deploy ML models quickly. It removes the heavy lifting from each step of the machine learning process.

Develop

SageMaker Studio IDE, notebooks, and preprocessing tools

Monitor

Model performance tracking and drift detection

Orchestrate

ML pipelines and workflow automation



Train

Distributed training with built-in or custom algorithms

Tune

Automated hyperparameter optimization and AutoML

Deploy

Real-time, batch, and serverless inference endpoints

SageMaker Studio: Unified ML Development

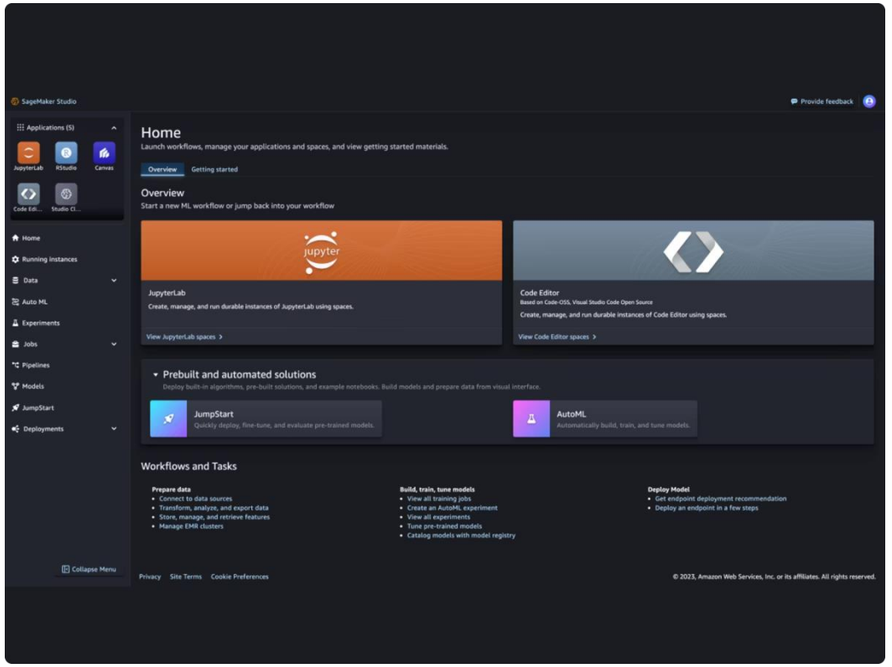
SageMaker Studio is a web-based IDE for machine learning that provides a single interface for all ML development steps. It's built on JupyterLab and includes integrated tools for every phase of the ML workflow.

Key Features

- Managed Jupyter notebooks with instant startup
- Visual workflow designer for ML pipelines
- Integrated data preparation with Data Wrangler
- Experiment tracking and model registry
- Collaborative workspace for teams
- One-click deployment to endpoints

Development Tools

- **Data Wrangler:** Visual data prep with 300+ transforms
- **Feature Store:** Centralized feature repository
- **Debugger:** Real-time training insights
- **Clarify:** Bias detection and explainability
- **Autopilot:** Automated model building



Training Options in SageMaker

Built-in Algorithms

SageMaker provides 18+ optimized algorithms for common ML tasks: XGBoost, Linear Learner, DeepAR for forecasting, Object Detection, Image Classification, Semantic Segmentation, Seq2Seq, and more. These algorithms are pre-tuned for performance and automatically scale across multiple instances.

Advantage: No code required for training—just point to your data and configure hyperparameters. AWS maintains and updates these algorithms.

Bring Your Own Container (BYOC)

Package any ML framework or custom code in a Docker container. SageMaker will run your container on managed infrastructure with automatic scaling, distributed training, and spot instance support.

Advantage: Complete flexibility to use any framework (TensorFlow, PyTorch, scikit-learn, or custom code) while benefiting from SageMaker's infrastructure management.

Script Mode with Framework Containers

Write training scripts in TensorFlow, PyTorch, or other frameworks. SageMaker provides pre-built containers with these frameworks installed, so you just supply your training script.

Advantage: Balance between simplicity and customization—use familiar frameworks without managing Docker images.

Data Inputs and Storage Patterns

SageMaker training jobs pull data from various sources, with S3 being the primary storage layer. Understanding data input modes and storage patterns is crucial for optimizing training performance and costs.

01

Stage Data in S3

Store training datasets in S3 buckets organized by project and version. Use S3 lifecycle policies to manage storage costs.

02

Choose Input Mode

File mode: Downloads entire dataset to instance storage before training (default).

Pipe mode: Streams data directly from S3, reducing startup time and storage needs.

03

Configure Channels

Define input channels for training, validation, and test data. SageMaker makes these available to your training code via environment variables.

04

Optional: Feature Store

For production ML, use SageMaker Feature Store as a centralized repository for curated features, ensuring consistency between training and inference.

Cost Optimization: Spot Instances and Distributed Training

Managed Spot Training

SageMaker can use EC2 Spot Instances for training jobs, offering up to 90% cost savings compared to on-demand pricing. SageMaker automatically handles interruptions by checkpointing and resuming training.

- Enable with one parameter: `use_spot_instances=True`
- Set max wait time and max run time
- SageMaker handles checkpointing automatically for built-in algorithms
- Best for training jobs that tolerate interruptions

Typical savings: 70-90% for long-running training jobs

Distributed Training

Split training across multiple instances to reduce training time for large models and datasets. SageMaker supports data parallelism and model parallelism.

- **Data parallelism:** Split data across instances, each running full model
- **Model parallelism:** Split large models across instances when model doesn't fit in memory
- Automatic gradient synchronization
- Works with TensorFlow, PyTorch, and built-in algorithms

Trade-off: Faster training vs. higher instance costs

Deployment Options: Real-Time Inference

Real-time inference endpoints provide low-latency predictions for applications requiring immediate responses. SageMaker hosts models behind HTTPS endpoints with automatic scaling and load balancing.



Single-Model Endpoints

Deploy one model to an endpoint backed by one or more instances. Use auto-scaling policies to add/remove instances based on request volume or latency targets. Choose instance types based on model size and latency requirements.

Multi-Model Endpoints

Host multiple models on a single endpoint, ideal when you have hundreds of models serving similar workloads. Models are loaded into memory on-demand and cached. Reduces costs by sharing infrastructure across models.

Serverless Inference

Deploy models without provisioning infrastructure. AWS automatically scales capacity based on demand, scaling to zero when idle. Perfect for intermittent traffic patterns or new applications with unpredictable load.

Batch Transform for Offline Predictions

When you don't need real-time predictions, Batch Transform provides cost-effective inference for large datasets. It automatically provisions infrastructure, processes your data, and shuts down when complete.

1

Upload Input Data to S3

Organize your input data in S3 with consistent formatting (CSV, JSON, or image files).

2

Configure Batch Job

Specify input/output S3 locations, instance type and count, and batch size per request.

3

Execute Batch Transform

SageMaker provisions instances, processes data in parallel, and writes predictions to S3.

4

Automatic Cleanup

Infrastructure is automatically terminated after job completion—you only pay for actual processing time.

Use cases: Monthly churn predictions, daily inventory forecasting, periodic image classification, end-of-day transaction fraud scoring. **Cost advantage:** No need to keep endpoints running 24/7 for periodic scoring.

Async Inference for Long-Running Predictions

Async inference handles requests that take minutes to process, such as large document analysis or complex simulations. Requests are queued, processed asynchronously, and results are stored in S3.

Queue-Based Architecture

Requests are placed in an SQS queue, allowing the endpoint to scale based on queue depth. Clients receive immediate acknowledgment and poll for results.

Auto-Scaling to Zero

Unlike real-time endpoints, async endpoints can scale to zero instances when the queue is empty, significantly reducing costs for variable workloads.

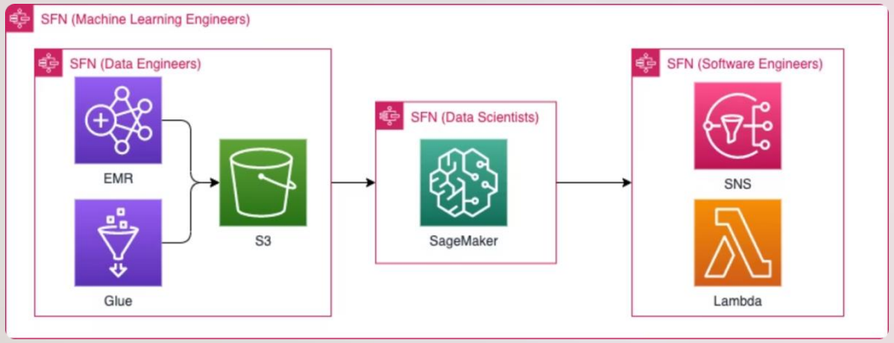
S3 Integration

Input data and predictions are stored in S3. Optionally configure SNS notifications when predictions are ready, enabling event-driven workflows.

Best for: Video analysis, large document processing, complex simulations, or any prediction taking 15 minutes or longer.

MLOps with SageMaker Pipelines

SageMaker Pipelines is a purpose-built CI/CD service for machine learning that automates and standardizes the ML workflow from data preparation through deployment.



Pipeline Components

- **Processing steps:** Data validation and feature engineering
- **Training steps:** Model training with parameter tracking
- **Evaluation steps:** Model quality assessment
- **Condition steps:** Conditional logic based on metrics
- **Model registration:** Automatic versioning in Model Registry
- **Deployment steps:** Automated model deployment

Benefits

- Reproducible workflows with versioned artifacts
- Experiment tracking and lineage
- Automated retraining triggers
- Parallel execution of independent steps
- Integration with EventBridge for scheduling
- Approval gates for production deployments

Pipelines are defined in Python using a directed acyclic graph (DAG) structure, similar to Airflow but optimized for ML workflows.

Cost Optimization Strategies

Training Costs

- Use Spot Instances for 70-90% savings on training
- Right-size instance types based on model requirements
- Use pipe mode to reduce EBS storage costs
- Enable SageMaker Debugger to catch inefficiencies early
- Share compute via multi-model training jobs

Inference Costs

- Use serverless inference for variable or low-volume traffic
- Choose batch transform over real-time for offline scoring
- Use async inference for long-running predictions
- Deploy multi-model endpoints to share infrastructure
- Configure auto-scaling policies to match actual demand

General Strategies

- Use S3 Intelligent-Tiering for training data
- Clean up unused endpoints and notebook instances
- Use AWS Budgets to set spending alerts
- Tag resources for cost allocation tracking
- Review Cost Explorer monthly for optimization opportunities

Security and Governance Best Practices

Security Controls

- **IAM roles:** Least privilege access for SageMaker execution roles and users
- **VPC configuration:** Run training jobs and endpoints in private subnets
- **Encryption:** Enable encryption at rest (KMS) and in transit (TLS)
- **VPC endpoints:** Private connectivity to S3 and other AWS services
- **Network isolation:** Disable internet access for sensitive workloads
- **Logging:** Enable CloudTrail and CloudWatch for audit trails

Governance Framework

- **Service quotas:** Set limits to prevent runaway costs
- **SCPs:** Organization-level policies to enforce standards
- **Budget alerts:** Proactive notifications before overspending
- **Resource tagging:** Consistent tagging for cost allocation and organization
- **Model approval:** Require manual approval before production deployment
- **Compliance:** Use AWS services that meet regulatory requirements (HIPAA, PCI, etc.)

Implement defense-in-depth: combine network isolation, encryption, access controls, and monitoring for comprehensive security. Use SageMaker's built-in security features rather than building custom solutions.

References

- https://docs.aws.amazon.com/sagemaker/
- https://aws.amazon.com/documentation-overview/sagemaker/

Thank You